

M/V William James - Major Overhauls of Two (2) 3512 Main Engines

Scope of Work

The USACE MVK Revetment Branch requires an inframe major overhaul of two (2) Caterpillar 3512 marine propulsion engines on the M/V William James. The contractor will be required to disassemble, make repairs, and reassemble onsite onboard the M/V William James. All aspects of the major overhauls shall be performed and completed in accordance with the manufacturer's specifications/maintenance manual for a Caterpillar 3512 marine propulsion engine. All parts to be replaced shall be new OEM Caterpillar parts or Caterpillar reconditioned parts where allowed by the Caterpillar service manual recommendations. All reconditioned parts shall meet Caterpillar specifications. All new or reconditioned parts replaced and any service required to replace said parts shall have no less than a one-year factory Caterpillar warranty on parts and labor. All work shall be completed by Caterpillar certified technicians.

Requirements

- Perform inframe major overhaul of two (2) Caterpillar 3512 marine propulsion engines
- Both 3512 marine propulsion engines located onboard M/V William James
 - Arrangement number: 2322505
 - Serial numbers: S2J00211 & S2J00214
- Perform overhauls in accordance with Caterpillar maintenance manual for major overhaul
- All parts shall be new or reconditioned (where specified in the manual)
- Part List is attached at the end of this scope of work, though some part numbers could have changed
- Replace the following parts as part of major overhauls in conjunction with attached parts list:
 - Turbo Cartridges
 - Oil Coolers
 - Oil Pumps
 - Lifters & Lifter Springs
 - Cylinder Heads
 - Fuel Injectors w/New Harnesses
 - Cylinder Packs
 - Fuel Transfer Pumps
 - Water Pump
 - Auxiliary Water Pump
 - Fuel Priming Pump
 - Front/Rear Seals
 - Main Bearings
 - Rod Bearings
 - Thrust Bearings
 - Camshaft Bearings
 - Rocker Arm Bearings

- Temperature Regulators
- Crankcase Breathers
- All Air, Fuel, and Oil filters
- Relief Valves
- Oil Duplex Check Valves
- Steel Spacer Plates
- Valve Bridges
- Rocker Arm Buttons & Rings
- Engine Dampers
- Turbo Inlet & Outlet Bellows
- All seals and gaskets to replace above listed parts
- Replace all engine oil with Caterpillar approved oil
- Replace all ELC coolant
- Inspect the following to acceptable Caterpillar tolerances and suggest for replacement if necessary
 - Camshafts
 - Crankshafts
- Perform the following as part of major overhauls
 - Clean & Test Aftercoolers
 - Test run engines after completion
- Replace oil pan groups for each engine
 - Breakout chockfast mounting system to be able to remove and install oil pan group
 - Replace oil pan group – parts list included at end of scope of work
 - Reinstall chockfast or other equivalent mounting system suitable for proper engine operation
 - Provide proper engine alignment
- Replace display in engine control room for both engines
 - Replace CMD Display Group including
 - Display drop harness
 - CMD Wiring Connection Kit
- Contractor shall provide all materials and labor necessary for all work in requirements

Core Charges

Any old parts that cannot be reused by Caterpillar and will require a core charge to be applied shall be figured at completion of the job. A modification request with quote shall be issued by the contractor to the USACE Vicksburg contracting office. If the core charges are deemed valid by the government and cost is reasonable, a contract modification shall be issued by the USACE Vicksburg contracting office.

Period and Location of Performance

All work shall be completed no later than June 30, 2026, and will be completed onboard the M/V William James at the USACE Vicksburg Harbor at 2577 Haining Rd, Vicksburg, MS, 39183.

Payment and Invoicing

Payment will only be given for work/requirements that are completed. The invoice(s) shall be sent to the following email addresses:

CEFC-B4invoices@usace.army.mil

Nathanael.j.jabour@usace.army.mil

Security Requirements

General security requirements and guidance: The security requirements described below apply to all contract personnel (including employees of the prime Contractor ("Contractor") and all subcontractor employees) supporting the performance requirements of this contract. The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed)). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training contributes to those efforts. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared to provide copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as "RAMs"), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures -- this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract

personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

Physical security and access control requirements: All contract personnel requiring physical access to a federal installation or facility shall comply with the access control procedures of that location. Contract personnel requiring unescorted access to meet contract performance requirements on a DoD installation in the US shall be vetted by the installation/facility Provost Marshal/Directorate of Emergency Services/Security Office using the National Crime Information Center-Interstate Identification Index (commonly referred to as “NCIC-III”) and Terrorist Screening Database (commonly referred to as “TSDB”). Contract personnel shall comply with all personal identity verification requirements specified in installation/facility policies and procedures. Contract personnel who do not meet requirements for unescorted access to USACE facilities shall coordinate escorted access with the Government representative, as needed. Contract personnel who receive keys, access cards, or lock combinations that provide access to government-owned property shall comply with key and lock control procedures of the RA.

Suspicious Activity Reporting training (e.g. iWATCH, CorpsWatch, or See Something, Say Something): All contract personnel shall receive initial and annual refresher training from the RA representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative or law enforcement entity.

Escorting in classified and/or sensitive areas: In accordance with applicable regulations, all contract personnel who do not possess the appropriate security clearance or access privileges will be escorted in areas where they may be exposed to classified information or operations, sensitive information or activities, or restricted areas.

Pre-screen candidates using E-Verify Program: Contractors shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (<https://www.e-verify.gov/>) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

Non-U.S. Citizens working on Government Contracts: Government Approval MUST be obtained PRIOR to any non-U.S. Citizen being permitted to work on a government contract. The Contractor must submit a letter requesting approval via the Contracting Officer/Contracting Officer

Representative to the Vicksburg District Security Officer. The letter must include the contract number, period of performance, location(s) of work performance and attach a copy of the individual's passport and visa/work authorization documents.

Parts List Per Engine

Line No	Part Number & Description	Quantity
	Engine Overhaul Parts List	
0001	422-9537 Explosion Relief Valve Kit	2
0002	4P-0710 Engine Air Filter	2
0003	115-3634 Air Cleaner Soot Filter	2
0004	1R-0726 Standard Efficiency Engine Oil Filter	4
0005	554-3009 97.5mm Long Check Valve	1
0006	5D-5957 104.37mm Inner Diameter Seal O Ring	3
0007	9M-2610 46.2mm Inside Diameter Seal-O-Ring	1
0008	5P-4892 3.53 x 59.92mm 75A FKM O-Ring	4
0009	6V-3907 3.53 x 183.74mm 75A FKM O-Ring	6
0010	6V-5048 1.83 x 8.92mm 75A FKM O-Ring	5
0011	6V-8001 85.32mm Inside Diameter Seal-O-Ring	1
0012	6V-8002 88.49mm Inside Diameter Seal-O-Ring	1
0013	2G-6303 30mm Internal Diameter Circular Seal	1
0014	214-7568 STOR O-Ring Seal (SAE 9/16-18)	2
0015	5P-5846 40.64mm Inside Diameter Seal-O-Ring	2
0016	112-3540 23.16mm Inside Diameter Seal-O-Ring	4
0017	272-2583 2.62 x 53.64mm 75A FKM O-Ring	2
0018	1R-0756 Element Assembly	7
0019	5D-5957 Seal O-ring	2
0020	162-3906 188.4cc Fuel Priming Pump	1
0021	9Y-8389 1.2mm Thick Fuel Pump Gasket	1
0022	238-5081 2.46 x 19.18mm 90A FKM O-Ring	1
0023	238-5080 2.21 x 16.36mm 90A FKM O-Ring	1
0024	4W-3027 Engine Crankcase Breather	3
0025	033-6031 3.53 x 69.44mm 75A FKM O-Ring	3
0026	0R-5519 Cat® Reman Oil Cooler	1
0027	475-0905 Crankshaft-Bearing KIT - Main	1
0028	569-7729 272.288mm Outer Crankshaft Seal	1
0029	569-7728 272.288mm Diameter Crankshaft Seal	1
0030	20R-3545 Cat® Reman Cylinder Head	12
0031	265-7537 97mm Long Floating Bridge Inlet	12
0032	265-7538 Steel Exhaust Floating Bridge	12
0033	519-7297 212.5mm Diameter Spacer Plate	12
0034	7N-3368 0.8mm Thick Cylinder Block Gasket	1
0035	8N-1991 70.28mm Maximum OD Plastic Retaining Ring	1
0036	230-1072 0.8mm Thick Gasket	1
0037	6J-2680 3.53 x 69.44mm 90A NBR O-Ring	1
0038	8T-1919 62.87mm Inside Diameter Seal-O-Ring	1
0039	6V-1454 72.39mm Inside Diameter Seal-O-Ring	2
0040	3S-9643 Rotating Shaft Lip Type Seal	2

0041	649-0847 Temperature Regulator	2
0042	428-9130 118.50mm Maximum Inner Diameter Gasket	20
0043	7L-6443 3/8"-16 x 2 1/4" Hex Head Bolt	80
0044	110-8314 3/8" Steel Flat Washer	160
0045	2N-2766 3/8"-16 Cone Lock Nut	80
0046	1W-1564 190mm Long Engine Oil Pan Suction Screen	1
0047	9L-1480 0.8mm Thick Prelubrication Lines Gasket	1
0048	10R-9224 Cat® Reman Cylinder Pack	12
0049	355-8509 Bearing-Connecting Rod	12
0050	0R-3230 Cat® Reman Engine Oil Pump	1
0051	298-6387 3.53 x 164.69mm 75A FKM O-Ring	1
0052	9M-4355 Steel Bolt Damper for Transmission Planetary	3
0053	20R-1272 Reman Electronic Fuel Injector	12
0054	1H-9696 3.53 x 44.04mm 75A NBR O-Ring	1
0055	10R-9838 Cat® Reman Fuel Transfer Pump	1
0056	10R-1669 Cat® Reman Water Pump	1
0057	3B-8453 18.42mm Inner Diameter Seal O Ring	1
0058	235-3546 3.53 x 234.54mm 75A FKM O-Ring	1
0059	20R-8813 Cat® Reman Auxiliary Water Pump	1
0060	5N-1940 1.6mm Thick Flange Gasket	1
0061	418-4163 Water Temperature Regulator	4
0062	6V-1943 3/8"-16 x 1 1/8" Hex Head Bolt	12
0063	6V-6753 3/8"-16 x 1 3/8" Hex Head Bolt	8
0064	5M-2894 3/8" ID Steel Flat Washer	20
0065	207-1390 96mm Overall Length Stainless Steel Bellows	2
0066	10R-2340 Cat® Reman Turbocharger Cartridge	2
0067	287-7207 236.02mm Inside Diameter Seal-O-Ring	2
0068	289-4079 4.5mm Thick Spiral Wound Turbine Housing Gasket	2
0069	7E-6508 Unit Injector Cap	12
0070	9X-0352 2 Pin Connector Receptacle	12
0071	132-6469 Cylinder Head Valve Mechanism Wiring Harness	12
0072	9F-7707 19mm Long Retainer Ring	36
0073	154-6389 14.73mm Outer Diameter Rocker Arm Button	36
0074	10R-3139 Cat® Reman Upgrade-to-New (UTN) Valve Lifter	12
0075	7N-4782 31.38mm Free Length Lifter Guide Spring	24
0076	20R-1626 Cat® Reman Valve Lifter	24
0077	599-7275 Single Cylinder Head Install Kit	12
0078	457-6112 Turbocharger Installation Gasket Kit	2
0079	306-2351 Main Water Pump Install Gasket Kit	1
0080	126-4175 EXPANSION TANK INSTALL KIT	1
0081	427-0371 Block & Oil Pan Gaskets Kit - Central Lower Structure	1
0082	426-3622 WATER LINES GROUP GASKETS KIT	1
0083	464-0329 Oil Cooler and Lines Gasket Kit	1
0084	464-2432 Aftercooler & Lines Install Kit	1

Oil Pan Parts List

0085	1W-1564 190mm Long Engine Oil Pan Suction Screen	1
0086	4B-2363 1 1/2-12-2A Thread Square-Drive Plug	2
0087	6F-4868 1.6mm Thick Oil Pan Cover Gasket	2
0088	7C-4715 80mm Inner Diameter Flange Connector	1
0089	7W-0129 Flange Adapter	1
0090	111-2081 Oil Pan Tube Housing	1
0091	114-2413 85.6mm Inner Diameter Deep Sump Oil Pan Tube	1
0092	149-9298 Oil Pan Drain Cover	1
0093	228-8926 3/8" Engine Oil Pan Adapter	1
0094	229-8875 TUBE-SUCTION	1
0095	229-8877 1909.42mm Long Oil Pan	1
0096	231-5867 88.9mm Diameter Saddle Clamp	4
0097	0L-1351 3/8 - 16 Thread Hex Head Bolt	5
0098	0S-1585 1/2"-13 x 1 1/2" Hex Head Bolt	2
0099	165-3928 3/8"-16 x 1 1/4" Hex Head Bolt	4
0100	165-3932 3/8"-16 x 2" Hex Head Bolt	2
0101	0S-1594 3/8-16 Hexagonal Head Bolt	2
0102	259-4598 3.53 x 120.24mm 75A FKM O-Ring	1
0103	1A-2029 3/8"-16 x 1 1/8" Hex Head Bolt	4
0104	1K-6872 1/2"-13 Cone Lock Nut	2
0105	259-4600 5.33 x 85.09mm 75A FKM O-Ring	2
0106	4B-3140 10mm Thick Two Bolt Mounting Cover	1
0107	5M-2894 3/8" ID Steel Flat Washer	11
0108	5P-7818 91.44mm Inner Diameter O-Ring Seal	1
0109	5P-8245 M12 ID Steel Flat Washer	6
0110	5P-8872 113.66mm Inside Diameter Seal-O-Ring	1
0111	6V-5101 107.32mm Inner Diameter Seal O Ring	1
0112	7B-3235 1/2-13 Thread Hex Head Bolt	2
0113	310-3954 5.33 x 123.19mm 75A FKM O-Ring	2
0114	8M-9024 10.2mm Inner Diameter Washer	2
0115	8T-6762 1/2-14 NPTF Thread Square Drive Plug	1
0116	9L-1480 0.8mm Thick Prelubrication Lines Gasket	1
0117	7E-2630 Gasket Oil	1
0118	7W-0129 Flange Adapter	1
0119	261-5615 Bolt Hex Head	18
0120	165-3928 Bolt Hex Head	2
0121	245-3804 Bolt Hex Head	38
0122	3P-0654 Seal O-Ring	1
0123	5M-2894 Washer	40
0124	6V-5101 Seal	1
0125	8M-9124 Washer	18

Display Additions

0127	CMD13 Display Group	1
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0128	Display Drop Harness	1
0129	CMD Wiring Connector Kit	1